Problem 6

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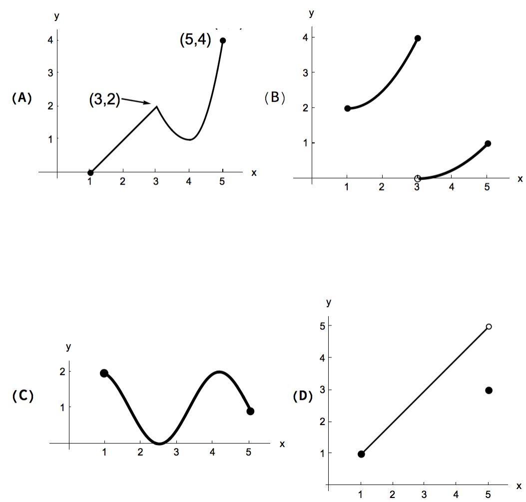
Problem 1 *Given the four functions on the interval* $[1, 5]$, *answer the questions below.*



(a) *List the functions that satisfy the hypothesis of the Extreme Value Theorem on* $[1, 5]$.

Explanation. The Extreme Value Theorem only requires that the function be continuous on $[1, 5]$. That means the functions in figures (A) and (C).

(b) *For each of the functions, determine if they have a global maximum, a global minimum, both, or neither.*

Explanation. The function in figures (A) and (C) satisfy the hypothesis of the Extreme Value Theorem, so they are guaranteed to have both a global maximum and a global minimum. The points where these global extrema are attained can be seen from the graph. (A) has a maximum at $x = 4$ and minimum at $x = 1$. (C) has a maximum at $x = 1$ and $x = 4$, and minimum around $x = \frac{5}{2}$.

The function in (B) has a maximum at $x = 3$, but no global minimum. The function in (D) has a minimum at $x = 1$, but no global maximum.

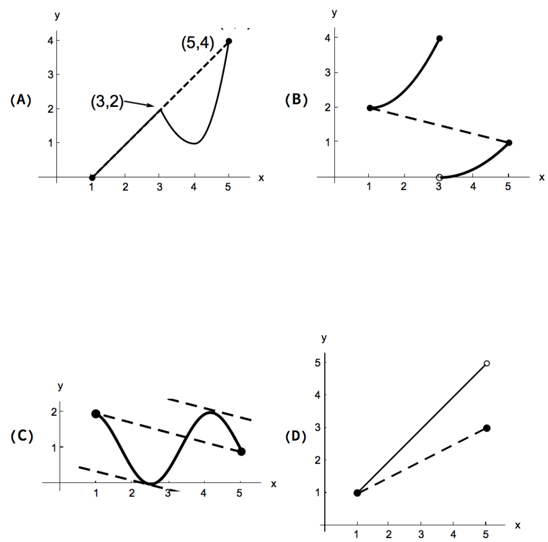
(c) *List the functions that satisfy the hypothesis of the Mean Value Theorem on* $[1, 5]$.

Explanation. Only the function in figure (C) satisfies the hypothesis of the Mean Value Theorem on $[1, 5]$. The function in figure (C) is **continuous** on $[1, 5]$ and **differentiable** on $(1, 5)$.
The function in figure (A) is **not differentiable** at $x = 3$, therefore it is **not differentiable** on $(1, 5)$.
The functions in figure (B) and (D) are **not continuous** on $[1, 5]$.

(d) *List the function (or functions) for which there exists a point* c *in* $(1, 5)$ *such that*

$$f'(c) = \frac{f(5) - f(1)}{5 - 1}$$

Explanation. Only the graphs (A) and (C). Explanation below. Look at these figures.



In figure(A), the secant line connecting the points $(1, f(1))$ and $(5, f(5))$ is also a tangent line to the curve at x, $1 < x < 3$.
(A): $\dfrac{f(5) - f(1)}{5 - 1} = \dfrac{1 - 5}{4} = -1$ and for any c in $(1, 5), f'(c) = -1$.
(C): $\dfrac{f(5) - f(1)}{5 - 1}$ is the slope of the secant line connecting the points $(1, f(1))$ and $(5, f(5))$. This line is parallel to two tangent lines shown in the figure (C). The lines have the same slope. The slope of either of a tangent line is $f'(s)$, where is $(s, f(s))$ is the point where the line touches the curve.
(B): The slope of the secant line connecting $(1, f(1))$ and $(5, f(5))$ is negative On the other hand $f'(s) > 0$, for $1 < s < 3$ and $3 < s < 5$. So,

$\dfrac{f(5)-f(1)}{5-1} = \dfrac{1-5}{4} = -\dfrac{1}{4} \neq f'(s)$, for all s where $f'(s)$ defined. (D): The slope of the secant line connecting $(1, f(1))$ and $(5, f(5))$ is given by $\dfrac{f(5)-f(1)}{5-1} = \dfrac{3-1}{4} = \dfrac{1}{2}$. On the other hand, on the interval $(1, 5)$, the graph of the function is a line whose slope is 1. Therefore, $f'(x) = 1$, for $1 < x < 5$.